Change making problem (tag: chng)

Training suite **Available memory: 512 MB. Maximum running time: ? s.**

You are given a collection of coins and a target value. Your task is to determine the smallest subset of coins whose values sum exactly to the target value. Each coin may be used at most once. Multiple coins may have the same denomination.

Input

The first line contains two integers, n ($1 \leq n \leq 5000$) and V ($1 \leq V \leq 20\,000$), denoting the number of coins and the target value, respectively.
The second line contains n integers representing the denominations of the coins. Each of these values is within range $[1, 20\,000]$.

Output

In the first line, output a single integer m, the minimum number of coins needed to obtain the value V. If it is impossible to obtain V using the given coins, output -1.
If $m \neq -1$, output a second line containing m integers: the denominations of the selected coins. These coins must form a subset of the input coins whose values sum to V.
If there are multiple optimal solutions, output any of them.

Subtasks

Subtask	Constraints
1	$n \leq 24$
2	$n \leq 48$
3	No additional constraints